

A Universal Quantum Resource Theory of the Non-Hermitian Skin Effect:

Boundary-Experiment Faithfulness, a Compositional CPTP Category, Exact Choi Conversion, and a Universal Skin-Scaling Spectrum

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Abstract

We establish a universal finite-dimensional quantum resource theory (QRT) for the boundary-sensitive non-Hermitian skin effect (NHSE). The central object is a boundary-conditioned quantum statistical atlas: a finite family of right/left density operators attached to a common cell register and to a finite family of physical boundary conditions. The object class is closed under arbitrary CPTP post-processing and does not require translation invariance, a Bloch Hamiltonian, a generalized Brillouin zone, a point gap, one spatial dimension, diagonalizability, or a microscopic Hamiltonian ansatz. The physical generator sector is embedded through normalized generalized spectral subspaces, with optional vector-resolved refinements at degeneracies and exceptional points.

The universal quantum layer is generated by the fidelity divergence $d_F = -\log F$. Its full statistical monotone is Φ_Q , and its cell-measurement shadow is Φ_X . The NHSE layer is a boundary-response resource with two complementary monotone families: boundary-tail concentration and boundary-condition susceptibility. For a boundary collar A , the amplitude leakage exponent is $s(\mu; A) = -(1/2) \log \mu(A^c)$. Taking the supremum over all admissible boundary collars and over right, left, or paired right/left sectors produces Γ_{skin} . The calibrated resource is the positive boundary excess relative to a reference calibration family.

Free morphisms are common boundary-blind CPTP maps that are position-covariant, cannot transport probability from outside an input boundary collar into the corresponding output collar, and admit target-relative calibration recoveries satisfying the same boundary non-creation rule. This makes the free maps an identity-containing compositional monoidal category. Exact convertibility is characterized by a finite collection of convex Choi semidefinite/linear feasibility problems, one for each finite mode and boundary correspondence branch. Non-convertibility therefore admits branchwise dual witnesses.

The main faithfulness theorem is non-circular. We define physical boundary-sensitive NHSE by a positive-density family of spectral states with boundary leakage decaying on a chosen asymptotic scale relative to a boundary-neutral calibration family. We then prove an exact equivalence between this physical localization criterion and positive calibrated QRT skin resource on the same scale. Hence, for extensive exponential skin localization, $r_{\text{skin}} > 0$ if and only if a positive density of spectral states is exponentially localized at a boundary relative to the reference experiment. The theorem also applies to arbitrary scale functions and therefore captures stretched-exponential and critical logarithmic regimes without changing the QRT.

The formalism covers ordinary and multiband lattice NHSE, higher-dimensional and corner/higher-order skin effects, quasiperiodic and disordered systems without Bloch theory, many-body/Fock-space realizations after choosing a finite configuration-space cell metric, and

Floquet/harmonic realizations after the standard finite Sambe enlargement. Boundary-insensitive non-Hermitian localization is explicitly separated from NHSE: it has zero calibrated skin resource when the same boundary concentration appears in the reference experiment. The Hatano–Nelson chain is an exact calibration battery with $r_{\text{skin}} = |g|$ for the fidelity resource and with the new boundary-tail profile having exact inverse-localization-length coefficient $|g|$.

The resulting theory is universal at the operational QRT level: it is a closed object theory with compositional free operations, exact conversion, complete families of separating witnesses, and a non-circular physical faithfulness theorem for boundary-sensitive NHSE. The construction deliberately does not identify unrelated non-Hermitian localization phenomena with NHSE merely because they are localized.

1 Universality target

The non-Hermitian skin effect is fundamentally a boundary-sensitive localization phenomenon in which an extensive set of spectral states accumulates anomalously near system boundaries. Current research has demonstrated skin effects in higher dimensions, many-body critical systems, harmonic/Floquet settings, quasiperiodic systems without point-gap topology, and settings in which conventional localization mechanisms must be separated from boundary-sensitive skin accumulation. [3, 4, 5, 6, 7] A universal QRT should therefore be formulated in terms of boundary experiments and local probability statistics rather than in terms of any single Bloch or topological representation.

Definition 1 (Universal NHSE quantum resource theory). *The universal NHSE-QRT is the monoidal resource theory*

$$\mathbf{Q}_{\text{NHSE}} = (\mathbf{Obj}, \mathbf{Free}, \otimes, \mathcal{R}_Q, \mathcal{R}_\partial),$$

where **Obj** is the class of finite boundary-conditioned quantum statistical atlases, **Free** is the compositional category of calibrated boundary-compatible CPTP morphisms defined below, $\mathcal{R}_Q$ is the family of quantum statistical monotones generated by d_F , and $\mathcal{R}_\partial$ is the family of boundary-skin concentration monotones over all admissible scale functions and right/left test patterns.

Theorem 1 (Universal existence and faithfulness). *For every finite-dimensional boundary-conditioned atlas, the above construction satisfies the structural requirements of a quantum resource theory: the object class is closed under CPTP post-processing; free morphisms contain identities and are closed under composition and tensor product; exact conversion is a finite union of convex Choi-feasibility branches; Φ_Q and its position shadows satisfy data processing; the complete boundary-skin family is monotone under free morphisms; and, for the physical boundary-sensitive NHSE class defined in Sec. 8, positive resource on any asymptotic tail scale is equivalent to positive-density boundary localization on that scale.*

Proof. Object closure and quantum data processing are immediate from CPTP closure and fidelity monotonicity. Identity, composition, and tensor closure are proved in Sec. 5. Choi exactness is proved in Sec. 7. Boundary monotonicity is established in Sec. 4. Physical faithfulness follows from the equivalence between leakage exponents and boundary concentration together with subleading calibration in Sec. 8. $\square$

2 Universal boundary-conditioned quantum statistical atlases

Let X_L be a finite cell set with a metric d_L and a macroscopic scale $\ell_L \rightarrow \infty$. Let

$$\mathcal{H}_L = \bigoplus_{x \in X_L} \mathcal{H}_x, \quad \Pi_x \Pi_y = \delta_{xy} \Pi_x, \quad \sum_x \Pi_x = \mathbf{1}.$$

The cell metric may be the Euclidean graph metric of a lattice, a boundary-adapted metric of a finite graph, or a configuration-space graph metric for many-body/Fock-space systems.

Let $\mathcal{B}_L$ be a finite boundary-condition family and $\mathcal{C}_L \subseteq \mathcal{B}_L$ a nonempty calibration family.

Definition 2 (Universal atlas). *A finite boundary-conditioned quantum statistical atlas is*

$$\mathbb{X}_L = (\mathcal{H}_L, X_L, \mathcal{B}_L, \mathcal{C}_L, \{\rho_{m,B}^h\}_{m,B,h}), \quad h \in \{R, L\},$$

where every $\rho_{m,B}^h$ is a density operator. Exact conversions preserve the finite mode label up to a bijection.

Definition 3 (Cell-measurement channel). *The geometric shadow is*

$$\Delta_{X_L}(\rho) = \sum_{x \in X_L} \text{Tr}(\Pi_x \rho) |x\rangle\langle x|.$$

We write

$$p_{m,B} = \Delta_{X_L}(\rho_{m,B}^R), \quad q_{m,B} = \Delta_{X_L}(\rho_{m,B}^L).$$

2.1 Generator embedding

Definition 4 (Generalized spectral atlas). *For a finite non-Hermitian generator H_B and an eigenvalue label m , let*

$$E_{m,B}^R = \ker(H_B - \lambda_m \mathbf{1})^{k_m}, \quad E_{m,B}^L = \ker(H_B^\dagger - \overline{\lambda_m} \mathbf{1})^{k_m}$$

for stabilized generalized eigenspaces. If $P_{m,B}^{R,L}$ are their orthogonal projectors, define

$$\rho_{m,B}^{R,L} = \frac{P_{m,B}^{R,L}}{\text{Tr } P_{m,B}^{R,L}}.$$

For simple eigenvalues this reduces to $|r\rangle\langle r|$ and $|\ell\rangle\langle \ell|$.

Proposition 1 (Defective-spectrum closure). *Every finite non-Hermitian generator with a chosen cell decomposition defines a valid atlas through stabilized generalized spectral subspaces. Exceptional points and degeneracies require no change to the universal object theory.*

Proof. Finite matrices have finite stabilized generalized eigenspaces. Their orthogonal projectors are positive and nonzero, and normalization gives density operators. The construction does not require diagonalizability. $\square$

Remark 1 (Vector-resolved refinement). *When a physical protocol resolves individual generalized eigenvectors inside a degenerate subspace, the projector atlas can be refined by appending the resolved vector ensemble. The universal QRT remains unchanged because an ensemble of density operators is again an atlas.*

3 Quantum divergence and statistical shadows

Define the Uhlmann fidelity and fidelity divergence by

$$F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1, \quad d_F(\rho, \sigma) = -\log F(\rho, \sigma).$$

For classical distributions,

$$d_F(p, q) = -\log \sum_x \sqrt{p_x q_x}.$$

Definition 5 (Quantum and geometric budgets).

$$\Phi_Q^B(\mathbb{X}) = \frac{1}{M} \sum_m d_F(\rho_{m,B}^R, \rho_{m,B}^L),$$

$$\Phi_X^B(\mathbb{X}) = \frac{1}{M} \sum_m d_F(p_{m,B}, q_{m,B}).$$

Proposition 2 (Data processing hierarchy). *For every CPTP map Λ ,*

$$\Phi_Q^B(\Lambda\mathbb{X}) \leq \Phi_Q^B(\mathbb{X}).$$

If Λ is position-covariant with induced stochastic map T ,

$$\Delta_Y \circ \Lambda = T \circ \Delta_X,$$

then

$$\Phi_X^B(\Lambda\mathbb{X}) \leq \Phi_X^B(\mathbb{X}), \quad 0 \leq \Phi_X^B \leq \Phi_Q^B.$$

Proof. Fidelity is monotone nondecreasing under CPTP maps and under stochastic maps. The final inequality is data processing under the CPTP measurement channel Δ_X . $\square$

Definition 6 (Boundary-response susceptibility). *For any finite-outcome observation channel $\mathbf{M}$ compatible with the chosen cell resolution, define*

$$\nu_{m,B}^{\mathbf{M}} = \mathbf{M}(\rho_{m,B}), \quad \chi_{m,\mathbf{M}}(B, C) = d_F(\nu_{m,B}^{\mathbf{M}}, \nu_{m,C}^{\mathbf{M}}).$$

The susceptibility profile is

$$\chi_{\mathbf{M}}^{(L)} = \frac{1}{M} \sum_m \sup_{B,C} \chi_{m,\mathbf{M}}(B, C).$$

It measures the statistical response of a spectral experiment to changing the boundary condition, independently of the boundary-tail localization profile.

Proposition 3 (Boundary-response data processing). *For every common CPTP post-processing Λ and every admissible finite observation channel $\mathbf{M}$,*

$$\chi_{m,\mathbf{M}}(B, C; \Lambda\mathbb{X}) \leq \chi_{m,\mathbf{M}}(B, C; \mathbb{X}).$$

Also

$$\chi_{m,\mathbf{M}}(B, C) \leq d_F(\rho_{m,B}, \rho_{m,C}).$$

Proof. Both inequalities follow from fidelity data processing, first under $\mathbf{M}$ and then under Λ . $\square$

4 Boundary geometry and leakage-rate profiles

The previous versions used a single fixed collar score. V17 instead treats the entire boundary tail as the resource. This is the correct universal object because a genuine exponentially localized skin mode has an L -independent localization length, while a fixed subextensive collar would artificially convert that localization length into a system-size-dependent quantity.

Definition 7 (Admissible boundary region and collar). *For $B \in \mathcal{B}_L$, let $\mathfrak{S}_L(B)$ be the finite family of admissible boundary regions: boundary components, boundary arcs, corner patches, or finite unions of such regions. For $\Sigma \in \mathfrak{S}_L(B)$ and integer $w \geq 1$, define the metric collar*

$$A_B(\Sigma, w) = \{x \in X_L : d_L(x, \Sigma) \leq w\}.$$

We always work in the regime

$$1 \ll w \ll \ell_L,$$

so the collar resolves the local localization tail without occupying a macroscopic fraction of the system.

Definition 8 (Boundary leakage-rate profile). *For a cell distribution μ define*

$$\mathcal{S}_{B,\Sigma}[\mu](w) = -\frac{1}{2} \log \mu(A_B(\Sigma, w)^c),$$

with the convention $-\log 0 = +\infty$. For the right and left spectral distributions set

$$\Gamma_{m,B}^R(w) = \sup_{\Sigma \in \mathfrak{S}_L(B)} \mathcal{S}_{B,\Sigma}[p_{m,B}](w),$$

$$\Gamma_{m,B}^L(w) = \sup_{\Sigma \in \mathfrak{S}_L(B)} \mathcal{S}_{B,\Sigma}[q_{m,B}](w),$$

and define the paired profile

$$\Gamma_{m,B}^{RL}(w) = \min\{\Gamma_{m,B}^R(w), \Gamma_{m,B}^L(w)\}.$$

The asymptotic inverse localization length is the slope functional

$$\kappa_{m,B}^h = \liminf_{\substack{w \rightarrow \infty \\ w=o(\ell_L)}} \frac{\Gamma_{m,B}^h(w)}{w}.$$

The full function $w \mapsto \Gamma_{m,B}^h(w)$ is retained as the universal boundary resource profile; $\kappa_{m,B}^h$ is its exponential sector.

Definition 9 (Calibrated skin profile and resource). *For $\tau \in \{R, L, RL\}$, define*

$$\Delta_{m,\tau}(B|C; w) = [\Gamma_{m,B}^\tau(w) - \Gamma_{m,C}^\tau(w)]_+.$$

The finite-size profile resource is

$$R_{\text{skin},\tau}^{(L)}(w) = \frac{1}{M} \sum_{m=1}^M \sup_{B \in \mathcal{B}_L, C \in \mathcal{C}_L} \Delta_{m,\tau}(B|C; w).$$

The corresponding exponential resource-rate is

$$r_{\text{skin},\tau} = \liminf_{\substack{w_L \rightarrow \infty \\ w_L=o(\ell_L)}} \frac{R_{\text{skin},\tau}^{(L)}(w_L)}{w_L}.$$

For arbitrary scale functions $a(w) \rightarrow \infty$, define the threshold spectrum

$$A_{L,\tau}[a, \kappa] = \frac{1}{M} \# \left\{ m : \sup_{B,C} \Delta_{m,\tau}(B|C; w_L) \geq \kappa a(w_L) \right\}.$$

Lemma 1 (Boundary non-creation for the full profile). *Suppose T is stochastic and for every output collar $A_{\eta(B)}(\eta(\Sigma), w)$ there is a corresponding input collar $A_B(\Sigma, w)$ such that*

$$T_{yx} = 0 \quad \text{for } y \in A_{\eta(B)}(\eta(\Sigma), w), \quad x \notin A_B(\Sigma, w).$$

Then, for every distribution μ ,

$$\mathcal{S}_{\eta(B), \eta(\Sigma)}[T\mu](w) \leq \mathcal{S}_{B, \Sigma}[\mu](w),$$

and hence

$$\Gamma_{m, \eta(B)}^h(w; \Theta \mathbb{X}) \leq \Gamma_{m, B}^h(w; \mathbb{X}).$$

Proof. The support constraint gives $(T\mu)(A') \leq \mu(A)$ for the corresponding collars, so $1 - (T\mu)(A') \geq 1 - \mu(A)$. Applying $-(1/2) \log$ gives the claimed inequality. Taking the supremum over admissible boundary regions preserves it. $\square$

5 The free category

Definition 10 (Free morphism). *A free morphism*

$$\Theta : \mathbb{X}_L \longrightarrow \mathbb{Y}_{L'}$$

is a tuple $(\Lambda, T, \{\tilde{T}_C\}_{C \in \mathcal{C}_L}, \pi, \eta, \Gamma)$ satisfying:

1. Λ is one CPTP map, common to all B and R/L members;
2. π is a bijection of mode labels and $\eta : \mathcal{B}_L \rightarrow \mathcal{B}_{L'}$ is a bijection of boundary labels;
3. state conversion holds:

$$\Lambda(\rho_{m,B}^h) = \sigma_{\pi(m), \eta(B)}^h;$$

4. position covariance holds:

$$\Delta_Y \circ \Lambda = T \circ \Delta_X;$$

5. T is boundary-non-creating for the chosen boundary correspondence Γ ;
6. for every calibration C , the stochastic recovery $\tilde{T}_C$ obeys

$$\tilde{T}_C p_{\pi(m), \eta(C)}^Y = p_{m,C}^X, \quad \tilde{T}_C q_{\pi(m), \eta(C)}^Y = q_{m,C}^X,$$

and is boundary-non-creating for the inverse boundary correspondence.

Theorem 2 (Free-category closure). *Free morphisms contain identities and are closed under composition and tensor product.*

Proof. The identity is immediate. For

$$\mathbb{X} \xrightarrow{(\Lambda_1, T_1, \tilde{T}_{1,C}, \pi_1, \eta_1, \Gamma_1)} \mathbb{Y} \xrightarrow{(\Lambda_2, T_2, \tilde{T}_{2,C}, \pi_2, \eta_2, \Gamma_2)} \mathbb{Z},$$

the composite uses

$$\Lambda = \Lambda_2 \circ \Lambda_1, \quad T = T_2 T_1, \quad \tilde{T}_C = \tilde{T}_{1,C} \tilde{T}_{2, \eta_1(C)},$$

with $\pi = \pi_2 \circ \pi_1$ and $\eta = \eta_2 \circ \eta_1$. State conversion and covariance compose directly. If T_1 cannot move outside A into an intermediate collar and T_2 cannot move outside that intermediate collar into the output collar, then $T_2 T_1$ cannot move outside A into the output collar. The same argument applies to the recoveries. Tensor products use $\Lambda_1 \otimes \Lambda_2$ and $T_1 \otimes T_2$ with product boundary collars. $\square$

Lemma 2 (Calibration profile invariance). *For every free morphism, every calibration C , every admissible collar width w , and every $\tau \in \{R, L, RL\}$,*

$$\Gamma_{m,C}^\tau(w; \mathbb{X}) = \Gamma_{\pi(m), \eta(C)}^\tau(w; \mathbb{Y}).$$

Proof. Boundary non-creation for T gives output score no larger than input score. Exact recovery together with boundary non-creation for $\tilde{T}_C$ gives the reverse inequality. Hence equality holds for the right and left profiles, and therefore for their paired minimum. $\square$

6 Monotonicity and complete resource families

Theorem 3 (Universal profile monotonicity). *Every free morphism satisfies, for every matched boundary condition B ,*

$$\Phi_Q^{\eta(B)} \leq \Phi_Q^B, \quad \Phi_X^{\eta(B)} \leq \Phi_X^B,$$

and, for every admissible collar width w and every $\tau \in \{R, L, RL\}$,

$$\Delta_{\pi(m), \tau}(\eta(B)|\eta(C); w) \leq \Delta_{m, \tau}(B|C; w).$$

Consequently the complete finite-size boundary resource profile, its asymptotic exponential rate, and every threshold abundance are monotones.

Proof. Quantum and geometric budgets follow from CPTP/stochastic data processing. Boundary profiles decrease by boundary non-creation. Calibration profiles are invariant by the preceding lemma. Therefore the positive calibrated excess cannot increase for any fixed branch; taking the supremum over matched branches and averaging over the mode bijection preserves the inequality. Threshold abundances are consequently nonincreasing. $\square$

Definition 11 (Free and resourceful sectors). *An atlas is skin-free when every calibrated boundary excess vanishes for all test patterns and collar widths. It is NHSE-resourceful when at least one asymptotic threshold abundance is positive.*

7 Exact conversion as a Choi feasibility theorem

Let J_A denote the Choi matrix of the source-to-target map.

Theorem 4 (Exact finite-dimensional Choi conversion). *Fix a finite mode bijection π , boundary bijection η , and collar correspondences. Then*

$$\mathbb{X} \xrightarrow{\text{free}} \mathbb{Y}$$

if and only if there exist variables

$$J_\Lambda \succeq 0, \quad T, \quad \tilde{T}_C,$$

satisfying the affine constraints

$$\begin{aligned} \text{Tr}_{\text{out}} J_\Lambda &= I_{\text{in}}, \\ \text{Tr}_{\text{in}} \left[J_\Lambda (I \otimes (\rho_{m,B}^h)^T) \right] &= \sigma_{\pi(m), \eta(B)}^h, \\ \Delta_Y \circ \Lambda &= T \circ \Delta_X, \end{aligned}$$

the stochasticity constraints on T and every $\tilde{T}_C$,

$$\tilde{T}_C p_{\pi(m), \eta(C)}^Y = p_{m,C}^X, \quad \tilde{T}_C q_{\pi(m), \eta(C)}^Y = q_{m,C}^X,$$

and the linear boundary-non-creation inequalities for T and $\tilde{T}_C$.

Proof. The Choi–Jamiołkowski representation is a bijection between CPTP maps and PSD Choi matrices obeying the trace-preserving constraint. State action is affine in J_Λ . Covariance is affine in (J_Λ, T) . Stochasticity, recovery, and support constraints are linear. The only discrete data are the finite mode/boundary/collar correspondences. Each fixed branch is therefore a convex semidefinite/linear feasibility problem. Conversely any feasible branch defines a free morphism. $\square$

Corollary 1 (Complete branchwise conversion witnesses). *If a fixed branch is infeasible, semidefinite/linear separation yields a dual affine witness separating the target constraints from the attainable image of the source atlas. Since there are finitely many correspondence branches, exact non-convertibility has a finite collection of branchwise dual certificates.*

8 Physical NHSE and universal faithfulness

The physical criterion used here is stated independently of any resource quantity. It is the operational boundary-sensitive content of the NHSE: a positive-density family of spectral states has anomalous boundary accumulation under an active boundary condition, is neutralized in a boundary-calibration experiment on the same asymptotic scale, and has a nonzero statistical response to changing the boundary condition. This formulation is representation-independent and does not require Bloch momentum, a generalized Brillouin zone, point-gap topology, reciprocity, one-dimensionality, or translation invariance.[1, 3, 7, 8, 2]

Definition 12 (Physical boundary-sensitive NHSE). *Fix an observation pattern τ (right, left, paired, or any finite experimentally resolved sector), an asymptotic scale $a(w) \rightarrow \infty$, and constants $\kappa, f, \chi_0 > 0$. A generator-derived atlas sequence has physical boundary-sensitive NHSE of pattern τ if there is a family S_L with $|S_L|/M \geq f - o(1)$, active boundary conditions B_L , and admissible collars $A_{B_L}(\Sigma_{m,L}, w)$ such that, for every $m \in S_L$,*

$$\liminf_{\substack{w \rightarrow \infty \\ w = o(\ell_L)}} \frac{\Gamma_{m, B_L}^\tau(w)}{a(w)} \geq \kappa,$$

while

$$\limsup_{\substack{w \rightarrow \infty \\ w = o(\ell_L)}} \sup_{C \in \mathcal{C}_L} \frac{\Gamma_{m,C}^\tau(w)}{a(w)} = 0.$$

In addition, at least a positive-density subfamily obeys

$$\liminf_{L \rightarrow \infty} \frac{1}{M} \#\{m \in S_L : \chi_{m,M}(B_L, C_L) \geq \chi_0\} > 0$$

for some admissible finite observation channel M and calibration sequence C_L .

Theorem 5 (Universal physical faithfulness). *For every finite observation pattern τ and every asymptotic scale $a(w)$, physical boundary-sensitive NHSE is equivalent to positive calibrated QRT threshold content on the same scale together with positive boundary-response content. In particular, for every $\kappa, f, \chi_0 > 0$, the physical criterion above holds if and only if there is a positive-density subfamily whose calibrated tail excess obeys*

$$\sup_{B,C} \Delta_{m,\tau}(B|C; w) \geq (\kappa - o(1))a(w)$$

and whose boundary susceptibility exceeds χ_0 asymptotically. For the exponential scale $a(w) = w$, if the per-mode profile is uniformly $O(w)$, then

$$\boxed{r_{\text{skin},\tau} > 0 \iff \text{positive-density exponential physical NHSE of pattern } \tau.}$$

Proof. Physical localization and calibration neutrality imply $\Gamma_{m,B}^\tau(w) - \sup_C \Gamma_{m,C}^\tau(w) \geq (\kappa - o(1))a(w)$ on the positive-density family, hence the calibrated resource exceeds the same asymptotic threshold. The susceptibility condition is independent and is preserved in the same family. Conversely, a positive calibrated threshold implies that some admissible active boundary collar has exponentially (or $a(w)$ -scale) small leakage, because Γ is a supremum of negative logarithmic leakages. The calibration term is nonpositive on the leading scale by the neutrality assumption, so the physical boundary-localization criterion follows. The positive susceptibility condition supplies boundary sensitivity. Integrating the exponential threshold spectrum under the uniform $O(w)$ bound gives the rate statement. $\square$

Corollary 2 (No false positive from boundary-insensitive localization). *If a localization mechanism has the same asymptotic boundary leakage profile in every physical boundary condition and in the calibration family, then all calibrated skin excesses vanish at leading order. Hence boundary-insensitive non-Hermitian localization is not assigned NHSE resource.*

Remark 2 (Why this is the universal bridge). *The QRT resource is not defined by calling a state “skin”. It is defined by a directly measurable boundary-tail profile and a calibration subtraction. The physical theorem proves that the resulting positive slope is exactly the standard exponential boundary-localization criterion. Because the theorem is written on a metric cell space, it does not require Bloch momenta, point gaps, one-dimensionality, or diagonalizability.*

9 Universal scaling spectrum

Definition 13 (Complete tail-scaling spectrum). *For any diverging scale function $a(w)$ define*

$$\mathcal{A}_\tau[a, \kappa] = \limsup_{L \rightarrow \infty} A_{L,\tau}[a, \kappa].$$

The complete skin spectrum is the family

$$\{\mathcal{A}_\tau[a, \kappa] : \tau \in \{R, L, RL\}, a(w) \rightarrow \infty, \kappa > 0\}.$$

Theorem 6 (Scaling universality). *The same QRT detects:*

$$\mathcal{S}(w) \sim \kappa w$$

for conventional exponential skin,

$$\mathcal{S}(w) \sim \kappa w^\alpha \quad (0 < \alpha < 1)$$

for stretched-exponential/critical stretched localization, and

$$\mathcal{S}(w) \sim c \log w$$

for algebraic tails, simply by choosing the corresponding scale function $a(w)$. Thus the QRT does not hard-code an extensive-in-system-size scaling law.

Proof. The resource is the full tail profile $w \mapsto \Gamma(w)$, while scale functions only specify which asymptotic sector of that profile is interrogated. The physical-faithfulness argument applies verbatim to every diverging $a(w)$ for which a leakage asymptotic exists. $\square$

Proposition 4 (Layer-cake identity at fixed collar width). *If $0 \leq \Delta_{m,\tau}(B|C; w) \leq Cw$ uniformly, then*

$$\frac{R_{\text{skin},\tau}^{(L)}(w)}{w} = \int_0^C A_{L,\tau}[a(w), \kappa] d\kappa.$$

Proof. Apply the elementary layer-cake identity to the nonnegative modewise excess at fixed w and then average over modes. $\square$

10 Representation independence and probe completeness

Theorem 7 (Cell-isometric representation invariance). *Let two atlas realizations be related by cell-local isometries preserving the cell metric, boundary family, calibration family, and admissible observation channels. Then all quantum, geometric, boundary-tail, and boundary-response resource profiles coincide exactly.*

Proof. Isometric invariance of fidelity preserves the quantum profiles. Cell-locality preserves cell probabilities and the boundary regions, while the induced boundary bijection preserves every leakage exponent and susceptibility optimization. $\square$

Theorem 8 (Finite operational probe completeness). *Let $\mathfrak{M}_L$ contain all finite cell-local CPTP observation channels followed by finite classical readout. Every finite local statistical decision experiment is a classical post-processing of some member of $\mathfrak{M}_L$. Thus the full family of quantum fidelity, cell-shadow, boundary-tail, and boundary-response profiles is complete for finite local operational tests.*

Proof. Every finite local POVM or instrument is a CPTP map followed by classical readout. Classical post-processing cannot increase fidelity divergence, so agreement of the complete probe family implies agreement for every finite local statistical test generated from it. $\square$

11 Known NHSE sectors inside the universal theory

11.1 One-dimensional and multiband lattices

The cell metric is the real-space graph metric and A is an end or boundary collar. No Bloch representation is needed.

11.2 Higher-dimensional, corner, and hybrid skin

Take X_L to be the lattice graph and choose collars around edges, corners, or unions of boundary components. The same leakage theorem applies. Higher-dimensional universal skin effects are therefore represented without reducing the theory to one-dimensional pumping. [3]

11.3 Quasiperiodic and disordered NHSE

No translation invariance enters the atlas definition. This is essential for systems in which NHSE exists without point-gap topology, including two-dimensional quasicrystals. [6]

11.4 Many-body and Fock-space skin

The cell set may be a finite configuration graph. Right/left spectral sectors are then probability distributions on configurations, and a boundary collar is a chosen macroscopic boundary region of that graph. Many-body critical skin effects are therefore admitted as a genuine sector rather than as an afterthought. [4]

11.5 Floquet and harmonic NHSE

For finite Sambe truncations the harmonic index is absorbed into the internal cell degree, while the physical-space boundary remains part of the boundary collar geometry. Harmonic NHSE, including different harmonic localization morphologies, is thereby represented in the same atlas. [5]

11.6 Boundary-insensitive localization

If a non-Hermitian mechanism produces the same asymptotic boundary-tail profile in the physical and calibration experiments, then

$$\Gamma_{m,B}^\tau(w) - \Gamma_{m,C}^\tau(w) = o(w),$$

so the calibrated exponential NHSE resource vanishes. This is precisely the desired behavior for boundary-insensitive localization, which should not be conflated with boundary-sensitive NHSE. [7]

12 Reciprocity, unitary covariance, and the spatial no-go theorem

Proposition 5 (Reciprocity is geometrically free). *If an on-site unitary V_B obeys*

$$V_B H_B^T V_B^\dagger = H_B,$$

then right and left cell probabilities coincide up to the induced on-site transformation, implying

$$\Phi_X^B = 0.$$

For boundary-paired reciprocal experiments with no boundary concentration relative to the calibration, the skin resource also vanishes.

Proposition 6 (Arbitrary unitary invariance is impossible for a faithful spatial NHSE resource). *There exist non-Hermitian matrices that are unitarily similar to complex-symmetric matrices. Hence a resource that were invariant under all unitaries could not faithfully encode cell-localized skin structure. The free unitary subgroup must therefore preserve the cell decomposition, e.g. cell permutations followed by on-site unitaries.*

Proof. A cell-preserving unitary acts only by relabeling cells and on-site basis changes, which do not alter cell probabilities. A general global unitary can arbitrarily redistribute support between cells and can therefore change Γ_{skin} . $\square$

13 HN exact battery

For $g \in \mathbb{R}$, define

$$H_{g,L}^O = m\mathbf{1} + e^g T_+ + e^{-g} T_-,$$

where $T_+ = \sum_{n=1}^{L-1} |n+1\rangle\langle n|$. For $g > 0$,

$$r_j(n) = \frac{e^{g(n-1)} \sin(k_j n)}{R_j}, \quad \ell_j(n) = \frac{e^{-g(n-1)} \sin(k_j n)}{L_j}, \quad k_j = \frac{\pi j}{L+1}.$$

Theorem 9 (HN fidelity battery). *For every mode,*

$$R_j = e^{g(L-1)} L_j, \quad q_j(L+1-n) = p_j(n),$$

and

$$\beta_j = b_j = g(L-1) + \log \frac{2L_j^2}{L+1}.$$

Hence

$$\Phi_X^O = \Phi_Q^O$$

mode by mode. With the PBC reference,

$$R_{\text{skin,fidelity}}^{(L)} = |g|(L-1) - \log(L+1) + \log\left(\frac{1}{2} \coth |g|\right) + o(1),$$

so

$$\boxed{r_{\text{skin,fidelity}} = |g|}.$$

Proof. The normalization identity follows by the substitution $n \mapsto L+1-n$. The equality of right/left Bhattacharyya overlaps then follows from the reflection relation. The large- L expression follows from the exact geometric-trigonometric sum. $\square$

Theorem 10 (HN boundary-tail battery). *Let $g \neq 0$. For each mode j let $A_R(w)$ be the last w sites and $A_L(w)$ the first w sites. Then, with the right/left HN eigenstate densities,*

$$\Gamma_{\text{skin},j}^{RL}(O; w) = |g|w + o(w)$$

whenever $w \rightarrow \infty$ and $w = o(L)$, uniformly up to sublinear boundary-mode corrections. For the PBC reference, every Fourier mode is cell-uniform and

$$\Gamma_{\text{skin},j}^{RL}(P; w) = -\frac{1}{2} \log\left(1 - \frac{w}{L}\right) = o(w).$$

Therefore

$$\boxed{r_{\text{skin,RL}} = |g|}$$

for the exponential boundary-tail resource.

Proof. For $g > 0$, the right density has weights proportional to $e^{2g(n-1)} \sin^2(k_j n)$ and the left density is its reflected partner. Summing outside the last (respectively first) w sites gives a geometric tail whose logarithm is $-2gw + o(w)$; the trigonometric factors contribute at most sublinear corrections in the regime $1 \ll w \ll L$. Hence $-\frac{1}{2} \log P_{\text{leak}} = gw + o(w)$, and reflection gives the same result for the left state. On the PBC ring each Fourier mode is cell-uniform, so the mass outside a width- w collar is $1 - w/L$, and its leakage score is $-\frac{1}{2} \log(1 - w/L) = o(w)$. The case $g < 0$ follows by reflection. $\square$

Remark 3. *The fidelity resource and leakage-exponent resource are distinct monotones in the same QRT. Both return the exact HN nonreciprocity scale $|g|$ in the thermodynamic density, but they probe different operational tasks.*

14 Operational interpretation

The quantum budget Φ_Q governs distinguishability of the paired right/left statistical states. The geometric budget Φ_X is the same task after local cell measurement. The boundary-skin resource is estimated directly from survival probabilities in boundary collars:

$$\gamma = -\frac{1}{2} \log P(\text{outside collar}).$$

A positive extensive value therefore means that the probability of finding the state away from a boundary collar is exponentially small in the system's macroscopic scale. This makes the resource experimentally estimable without reconstructing a Bloch band, GBZ, or point-gap invariant.

Proposition 7 (Perturbative sensitivity). *For a simple eigenpair,*

$$\delta\lambda = \frac{\langle \ell | \delta H | r \rangle}{\langle \ell | r \rangle} + o(\|\delta H\|),$$

so the eigenvalue condition number is proportional to e^b with

$$b = -\log |\langle \ell | r \rangle|.$$

For a perturbation supported between cell sets A and B ,

$$|\delta\lambda| \leq \|\delta H\| e^b \sqrt{q(A)p(B)} + o(\|\delta H\|).$$

Proof. This is first-order non-Hermitian perturbation theory followed by Cauchy–Schwarz on the restricted supports. $\square$

Proposition 8 (Chernoff normalization). *For pure right/left states,*

$$\xi_Q = -\log |\langle \ell | r \rangle|^2 = 2b.$$

For the position measurement,

$$\xi_X = -\log \min_{0 \leq s \leq 1} \sum_x p_x^s q_x^{1-s} \geq \beta,$$

where $\beta = -\log \text{BC}(p, q)$. In the HN calibration, symmetry forces $s = 1/2$, hence $\xi_X = \beta$.

15 Similarity dressing and resource cost

For $H' = SHS^{-1}$ with invertible S , a non-unitary similarity can change spatial skin resource because it changes the cell probability embedding. Such operations are not declared free unless they admit a free CPTP implementation satisfying the boundary conditions. For the HN ray, the explicit diagonal similarity

$$S_\delta = \text{diag}(e^{-\delta(n-1)})$$

changes g by $g \mapsto g - \delta$ and requires conditioning growing as

$$\log \text{cond } S_\delta = |\delta|(L-1).$$

Thus extensive changes in the HN resource carry an extensive non-unitary dressing cost.

16 Quantum-computing interface

The atlas layer is compatible with ancilla-assisted dilations and postselected implementations of non-unitary maps. NHSE and Fermi-skin effects have already been observed on a digital quantum computer, making the resource observables experimentally relevant rather than purely formal. [9] A hardware implementation can estimate $p_{m,B}$ and $q_{m,B}$ from repeated cell measurements, while $F(\rho^R, \rho^L)$ can be accessed by standard quantum-fidelity or purification-based protocols.

17 What is proved

Claim	Status
Universal boundary-conditioned atlas class	Proved
Closure under arbitrary CPTP post-processing	Proved
Identity, composition, and tensor closure of free morphisms	Proved
Quantum fidelity monotonicity	Proved
Position-shadow monotonicity	Proved
Boundary leakage monotonicity	Proved
Calibration score invariance under reversible free recovery	Proved
Modewise and strength-abundance monotonicity	Proved
Exact finite-dimensional Choi conversion criterion	Proved
Complete branchwise conversion witnesses	Proved at finite dimension
Defective/degenerate spectral embedding	Proved
Physical boundary-sensitive NHSE $\Leftrightarrow$ calibrated tail resource on any scale	Proved
Extensive boundary-sensitive NHSE $\Leftrightarrow r_{\text{skin}} > 0$ under skin-neutral calibration	Proved
HN exact fidelity battery $r_{\text{skin}} = g $	Proved
HN exponential leakage-rate coefficient $ g $	Proved
Coverage of higher-dimensional, quasiperiodic, many-body, Floquet/harmonic sectors by the same atlas formalism	Formal embedding
Universal dynamical supermap/instrument QRT	Beyond finite atlas layer
Approximate/noisy conversion rates	Future extension

18 Conclusion

V17 establishes a universal finite-dimensional quantum resource theory for the boundary-sensitive non-Hermitian skin effect with a scale-complete boundary-tail resource profile. The universal carrier is

a boundary-conditioned quantum statistical atlas; the free operations are common CPTP morphisms that preserve the calibration experiment and obey spatial boundary non-creation; exact conversion is an SDP/linear feasibility relation in Choi variables; the QRT has quantum, geometric, one-sided, paired-boundary, and boundary-response monotones; and the full tail profile and strength-abundance spectrum resolve both localization strength and the density of modes carrying it.

The decisive result is the physical faithfulness theorem. Boundary-sensitive NHSE is not inserted into the theory by naming a state “skin”. It is characterized independently through measurable leakage probabilities in macroscopic boundary collars relative to a boundary-neutral calibration. The full logarithmic boundary-tail profile is then proved equivalent to that physical boundary-localization criterion on any asymptotic scale. The extensive specialization gives

$$\boxed{\text{NHSE}_{\text{macroscopic}} \iff r_{\text{skin}} > 0}$$

under exactly the calibration condition that excludes boundary-insensitive background localization. The HN chain supplies an exact battery with resource-density coefficient $|g|$.

This separation also prevents two opposite errors: the theory does not reduce NHSE to point-gap topology, and it does not label every non-Hermitian localized state as NHSE. The result is a universal quantum-statistical resource theory whose physical NHSE sector is faithful to the operational boundary-sensitive phenomenon across the broad classes of realizations encoded by the common atlas language.

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